University of Idaho
Department of Computer Science
Coeur d'Alene

Lecture #2

Vision and Cameras



Garrett Wells

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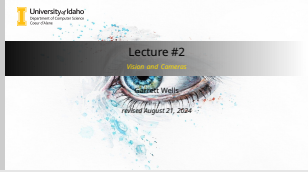


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Human Vision
- How Does the Eye Work?
- Vision and the Brain
- Other Factors
- Eye Conditions

Camera Origins
- Camera Obscura

Camera Operation
- Image Quality

Key Camera Concepts
- Camera Sensors
- Exposure Time a.k.a. Shutter Speed
- Focus
- Aperture
- Focal Length

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Stereoscopic Images

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Fun Human Vision Observations

Human Vision

What is the difference between human vision and machine vision?

- Human anatomy
- Frame rate

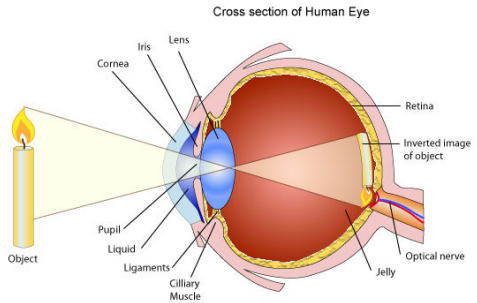


Figure: Human Eye Cross Section

Human Vision
What is the difference between human vision and machine vision?
- Human anatomy
- Frame rate
Figure: Human Eye Cross Section

1. Although human eye has components analogous to cameras, the deeper you look the more different they become.
2. Most obvious difference is that cameras operate in terms of frames per second. Human vision has a frame rate equivalency, but we don't see the world as a series of static images.

Human Vision

What is the difference between human vision and machine vision?

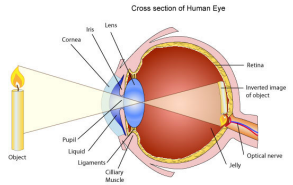


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Incident Light, light that falls on an object. Was historically measured in foot-candles (ft-c). One foot-candle is equivalent to the amount of light falling on a surface from a “standard candle” [1]

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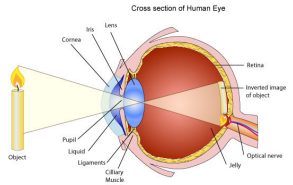


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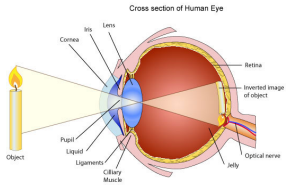


Figure: Human Eye Cross Section

- Light hits Cornea

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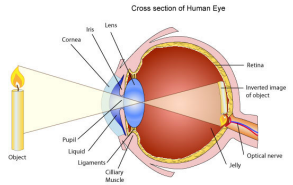


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- Light hits Cornea
- Light bends, hits Lens

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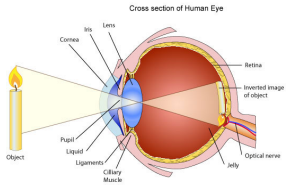


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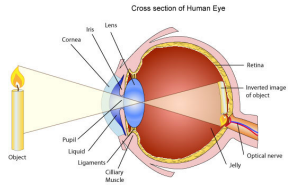


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- Light hits Cornea
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- Brain translates image (flips, etc)

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Human Vision

How does the brain process images?

1. Brain flips image
2. Visual cortex processes
   - Shape
   - Color
   - Movement, Location, Organization [2]

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Other factors…

- Iris Color
  - More melanin = light reflects less in eye

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  - Less melanin = less protected

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Human Vision

Other factors...

- Iris Color
  - Therefore, lighter eyes may be less comfortable in bright sunlight. [3]

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Human Vision

Red-Eye Effect

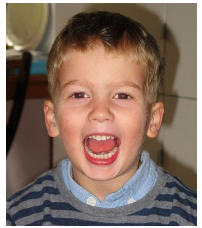


Figure: Classic Red-Eye Effect

- TLDR: light reflects off your retina, *"Light Reflex"*
- Illuminates blood supply of connective tissue creating a red color [4]
- Can be avoided by not using a flash.

Human Vision
Red-Eye Effect
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Red-Eye Effect



Figure: Contrasting Light Reflex

Cataracts and Retinoblastoma

Light reflex can produce white, yellow, or black retinas as seen in above, and can indicate conditions such as cataracts or retinoblastoma

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Human Vision

Eye Conditions...

Near-sightedness	*eye too long, focal point in front of retina*
Far-sightedness	*too short, focal point behind retina*
Cataracts	*cloudy lens*
Macular Degeneration	*degrading retina tissue*
Retinal Detachment	*aging, scar tissue, fluid buildup*
Keratoconus	*irregular cornea*
Astigmatism	*irregular cornea*

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Retinal Detachment aging, scar tissue, fluid buildup
Keratoconus irregular cornea
Astigmatism irregular cornea

A Brief History of Cameras

Camera Obscura

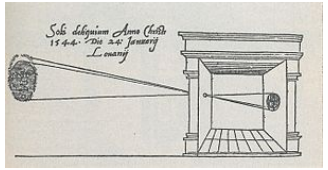


Figure: First published picture of Camera Obscura, Gemma Fisius, 1545

Definition

Camera Obscura "A natural optical phenomenon named after its Latin translation, "dark room". It projects an inverted image of a scene from the other side of a screen or wall through a small aperture onto a surface opposite the opening."
Wikipedia, [5]

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A Brief History of Cameras

- 18th-19th Century: Experiments in Chemistry
  - 1777 - Carl Wilhelm Scheele, Silver Chloride
  - 1826 - Joseph Nicephore Niepce, bitumen(petroleum)/“Heliography”
  - 1837 - Louis-Jacques-Mande Daguerre, “Daguerreotype”
  - 1885 - George Eastman, Paper Film

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1. Bitumen is a constituent of petroleum (see Wikipedia), hardens when exposed to sunlight and becomes insoluble. Impractical due to the hours required to harden.
2. Daguerreotype emerged from collaboration with Niepce, exposure time reduced to 5-30 min.
3. By 1889, Eastman was using celluloid in “Kodak” cameras. Celluloid film made it possible to make “motion pictures”.
4. These are all light sensitive materials, so once exposed, must be protected from further exposure until “developed”.
5. Development is the process of chemically treating the photosensitive material to make it stable.

A Brief History of Cameras

Digital Imaging...

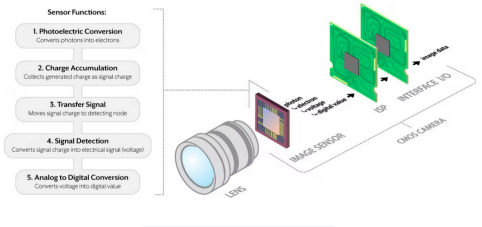


Figure: Caption

- 1969 - Willard S. Boyle & George E. Smith, Bell Labs, Produce first CCD image sensor
- 1989/90 - First digital handheld cameras commercially available

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Cameras

How do modern cameras work?

1. Light hits object.

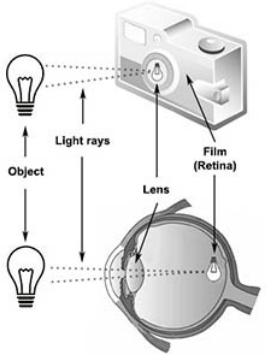


Figure: Eye vs. Camera

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Cameras

How do modern cameras work?

1. Light hits object.
2. Light reflects, striking camera lens.

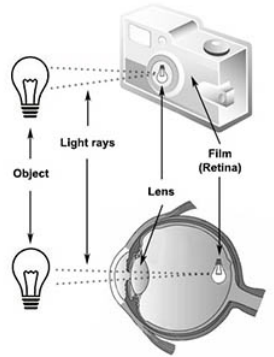


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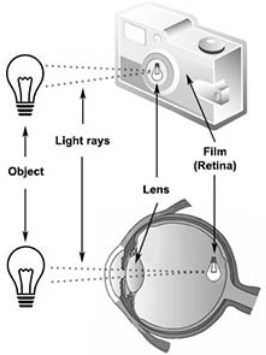


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1. Because light (sometimes) travels in straight lines, image is rendered upside down.

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    - film uses chemical reactions as an **analog** sensor
    - digital cameras use small sensors which produce voltage when activated by light

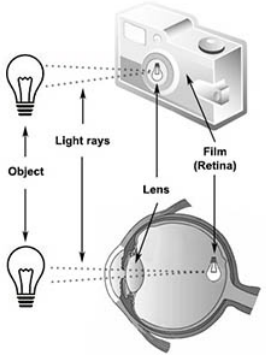


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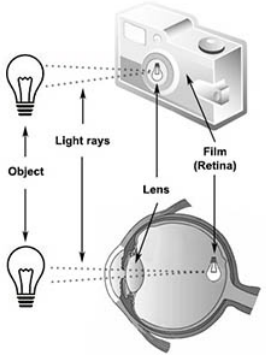


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Image Quality

How do we talk about image quality?

1. Focus
2. Exposure
3. Resolution

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How do we talk about image quality?
1. Focus
2. Exposure
3. Resolution

1. **Focus:** Is the lens focused properly to show fine texture/lines?
2. **Exposure:** Is there enough light for the sensor to see color and detail?
3. **Resolution:** The trickiest one to judge. More photoreceptors, more pixels, more detail? Generally, yes. Impacted by many variables. Always be aware that in MV it is possible to have *too many pixels to process in the time requirement!*

Key Aspects of Cameras & Lenses

Concepts to be familiar with.

- Camera Body
  - Sensor - *type, sensitivity, etc.*
  - Shutter
  - Interface - *USB, Ethernet, Memory Card*

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  - Focal Length
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Sensors

- Architecture
- Resolution
- Sensitivity
  1. set as ISO values
  2. higher sensitivity correlates to prevalence of “noise”

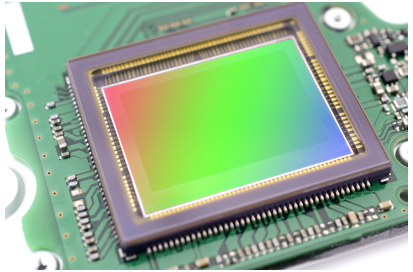


Figure: Example Sensor

1. Architecture: How is the light measured (CMOS/CCD)? What size is the sensor? How large are the pixels?
2. Resolution: May allow more detail, but also requires more processing time.

Sensors
- Architecture
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Exposure Time *a.k.a.* Shutter Speed

What is it and why do we care?

Exposure Time *a.k.a.* Shutter Speed

1. The time that the shutter is “open” allowing the sensor to measure the light striking it. Values are expressed in seconds.

Alternatively, shutter speed determines how many time steps or states in the scene contribute to the image formed on the sensor.

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1. The less light available, the longer the exposure time required for the sensor to “see” details in the scene.

Exposure Time *a.k.a.* Shutter Speed

Motion Blur…

- If the shutter is open for a period of time longer than it takes for the subject to cross the sensor, the subject will be blurred, or not even captured.

Exposure Time *a.k.a.* Shutter Speed
Motion Blur…
- If the shutter is open for a period of time longer than it takes for the subject to cross the sensor, the subject will be blurred, or not even captured.

1. Answers the question: *"If it's dark, why not just leave the shutter open longer?"*
2. Also, for real time applications(and those based on video), you may not be able to control the exposure time.

Exposure Time *a.k.a.* Shutter Speed

Final Observations: Light, F-Stops…

- Shutter speed values correlate to F-stops
- Video is quantized in *frames per second (fps)*

Exposure Time *a.k.a.* Shutter Speed
Final Observations: Light, F-Stops…
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- Video is quantized in *frames per second (fps)*

1. FPS is fixed in video: this means you don't get to change shutter speed on video cameras.
2. Can still change light, maybe sensor sensitivity

Focus

What is focus?

Definition

An image is considered “in focus” when the desired subject(s) are rendered with the maximum possible detail and clarity.

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Figure: Stages of Focus

Definition

An image is considered “in focus” when the desired subject(s) are rendered with the maximum possible detail and clarity.

1. Light is focused as points of light. In *“critical focus”* these points are theoretically zero radius
2. In reality, critical focus is almost impossible to achieve.
3. Thus the result are dots of light on the sensor referred to as *“Circles of Confusion”*.
4. The further the object is from the focal plane, the larger the circle of confusion rendered on the sensor.

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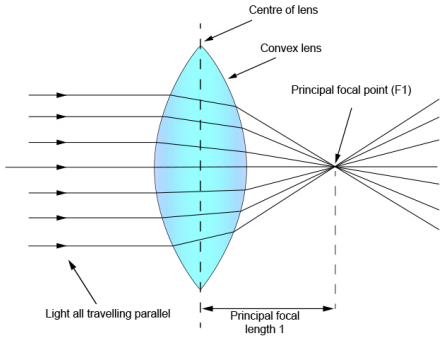


Figure: Convex Lenses and Focal Points

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Human Focus vs. Camera Focus

- Human eyes focus by changing shape of lens
- Cameras focus by changing lens distance from sensor

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Aperture

What is aperture?

1. Focus determined by changing distance
2. Aperture determines size of hole light passes through

Aperture
What is aperture?
1. Focus determined by changing distance
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1. Lateral translation of lens along central axis.
2. Aperture is another change *made to the lens*.

Aperture

What is aperture?



Figure: Half-stop Aperture Increments

Aperture
What is aperture?
f/1.4 f/2.0 f/2.8 f/4.0 f/5.6 f/8.0
Figure: Half-stop Aperture Increments

1. Images are of a large camera lens. It is much harder/impossible to observe this mechanism in your phone camera.
2. Aperture is closed by moving “Aperture Blades”.
3. A human’s pupil serves as the aperture mechanism of the eye.
4. The number of blades determines the shape of the opening(often geometric). Larger aperture closer to perfect circle when seen by human eye.
5. Changing the shape of the aperture even changes the shape of the dots of light on the sensor.

Aperture

What is aperture?

$$f\text{-}number = \frac{focal\ length\ (mm)}{aperture\ diameter\ (mm)}$$

$$aperture\ diameter\ (mm) = \frac{focal\ length\ (mm)}{f\text{-}number}$$

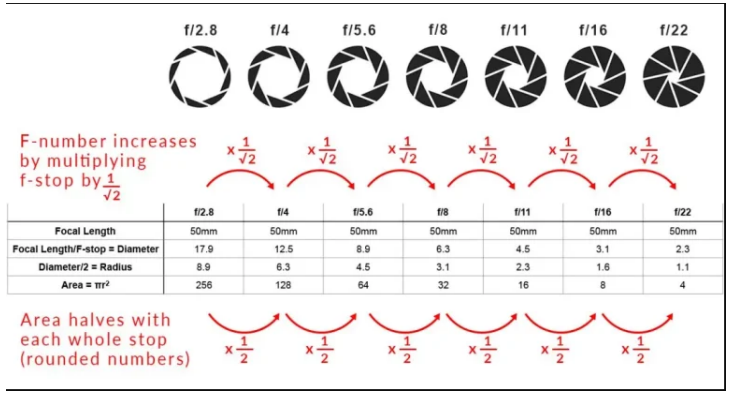
Figure: F-Stop Formula

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1. Lens invariant method of measuring relative light
2. Doesn't account for light lost in glass, etc.
3. Does allow for recording camera parameters to translate between camera setups, *i.e.* lenses, camera sensor, *etc*.

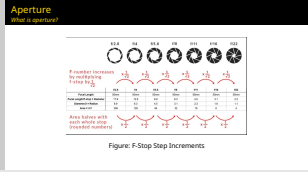
Aperture

What is aperture?



	f/2.8	f/4	f/5.6	f/8	f/11	f/16	f/22
Focal Length	50mm	50mm	50mm	50mm	50mm	50mm	50mm
Focal Length/F-stop = Diameter	17.9	12.5	8.9	6.3	4.5	3.1	2.3
Diameter/2 = Radius	8.9	6.3	4.5	3.1	2.3	1.6	1.1
Area = πr^2	256	128	64	32	16	8	4

Figure: F-Stop Step Increments



1. **Note:** $F \times (1/\sqrt{2}) =$ decrease by one stop

Aperture

What is aperture?

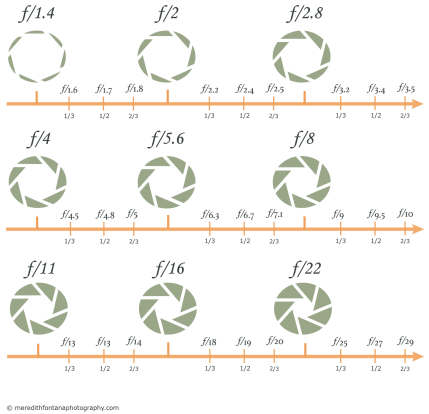
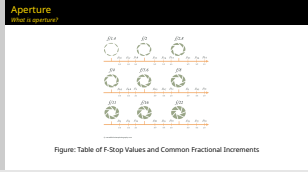


Figure: Table of F-Stop Values and Common Fractional Increments



1. F-stops can be broken in to smaller increments, but this is mostly of use for exposure compensation in photography.

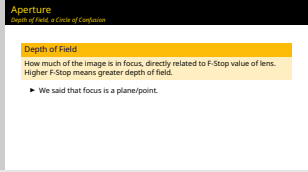
Aperture

Depth of Field, a Circle of Confusion

Depth of Field

How much of the image is in focus, directly related to F-Stop value of lens. Higher F-Stop means greater depth of field.

- We said that focus is a plane/point.



1. "Acceptable Sharpness" is vague. In most use cases it is based on the perception of the person viewing the image.
2. In machine vision applications, you may need to be aware of this because it relates to how much subject detail is available to the camera.
3. You may be able to numerically evaluate the sharpness.

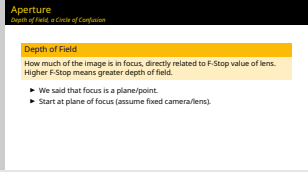
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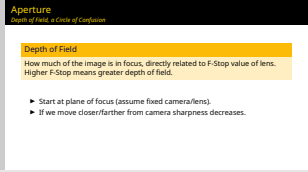
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- If we move closer/farther from camera sharpness decreases.



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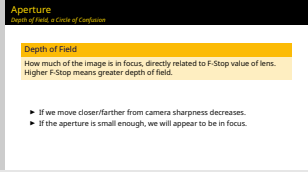
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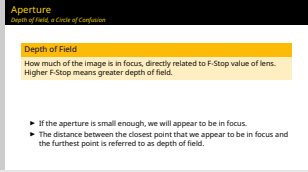
Aperture

Depth of Field, a Circle of Confusion

Depth of Field

How much of the image is in focus, directly related to F-Stop value of lens. Higher F-Stop means greater depth of field.

- If the aperture is small enough, we will appear to be in focus.
- The distance between the closest point that we appear to be in focus and the furthest point is referred to as depth of field.



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Focal Length

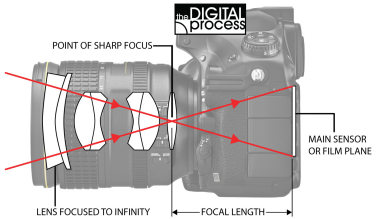


Figure: Focal Length Diagram

Definition

Focal Length Simplistically, the distance from where light converges in the lens to the sensor.

Focal Length
Figure: Focal Length Diagram
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Focal Length

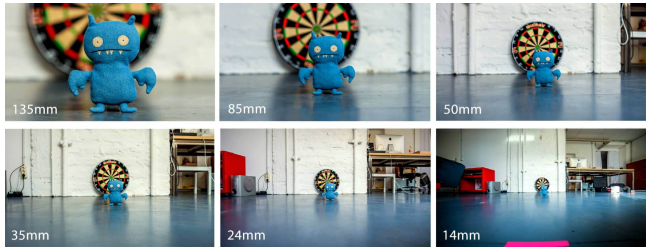


Figure: Focal Lengths Compared

Definition

Focal Length Simplistically, the distance from where light converges in the lens to the sensor.

Focal Length
Figure: Focal Lengths Compared
Definition
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Minimum Focal Distance

Definition

The minimum distance at which a lens is still able to focus on a subject.

Minimum Focal Distance
Definition
The minimum distance at which a lens is still able to focus on a subject.

1. It is an unpleasant surprise to invest in a camera with features you don't need, just to find that the camera needs to be placed 3-5ft from the subject.
2. Some cameras are permanently fixed with their focus at point maximizing depth of field.

Color

All the light we cannot see…

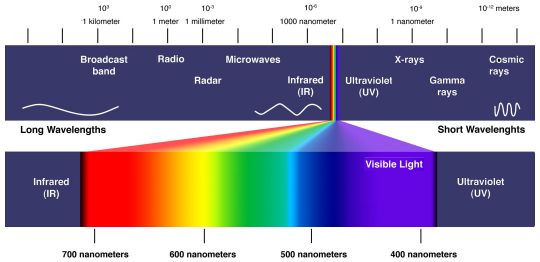
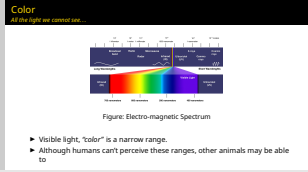


Figure: Electro-magnetic Spectrum

- Visible light, *“color”* is a narrow range.
- Although humans can’t perceive these ranges, other animals may be able to



1. **Note:** this is where machine vision is actually *better* than human vision. It can be tuned, filtered.

Stereoscopic/Binocular Vision

Too shallow? Take a second look.



Figure: Stereoscope

Definition

Stereoscope "optical instrument using two slightly differing pictures to illustrate binocular vision," 1838, coined by inventor Professor Charles Wheatstone (1802-1875)
from Etymology Online

Stereoscopic/Binocular Vision
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Figure: Stereoscope
Definition
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1. Produces sensation of looking at a 3D image.

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Figure: Stereoscopic Image Capture, Film

Definition

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Figure: Stereoscopic Image Capture, Film
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1. Unfortunately, rather crude. No real-time feedback.

Stereoscopic/Binocular Vision

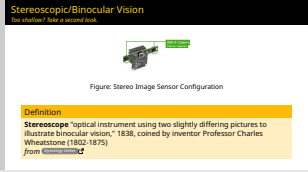
Too shallow? Take a second look.



Figure: Stereo Image Sensor Configuration

Definition

Stereoscope "optical instrument using two slightly differing pictures to illustrate binocular vision," 1838, coined by inventor Professor Charles Wheatstone (1802-1875)
from Etymology Online



1. Data published as a *"point-cloud"*, a 2D matrix of distances. One measurement per pixel.
2. Uses intersection of line-of-sight and disparity in images to calculate relative disparity of sensors.
3. This is then used to calculate depth. Possible to do this in OpenCV. https://vaibhavnandwani.medium.com/understanding-the-depth-how-stereo-cameras-see-our-world-in-3d-bd65a83ef5cc

Why Human Vision Is a Cheat

- **Saccade** Eyes rapidly jump between multiple objects of interest.
- Low light/Night Vision
- Object Recognition

Why Human Vision Is a Cheat
- **Saccade** Eyes rapidly jump between multiple objects of interest.
- Low light/Night Vision
- Object Recognition

1. Saccade movements make it possible for us to change focus rapidly, and build 3D map of a scene.
2. Vision becomes somewhat blurred during saccade movement.
3. Human vision in low light far exceeds camera performance. Cameras require long exposures to reveal anything more than black.
   May even need to stack multiple exposures to see color.
   Humans on the other hand can see relatively well “instantaneously”.
4. Humans are capable of recognizing objects from visual input. Amazingly powerful, key component of intelligence.
5. Scientists joke that object recognition of specific, meaningful objects indicates the existence of a hypothetical “Grandmother” cell/neuron responsible discerning “Grandma” from any other nice lady offering you cookies. Also known as the “Jennifer Aniston Neuron”.
   https://en.wikipedia.org/wiki/Grandmother$_c$*ell*

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[1] A. Adams and R. Baker, *The Camera*. Boston : New York Graphic Society, 1980, 228 pp., ISBN: 978-0-8212-1092-5. [Online]. Available: http://archive.org/details/cameraadam00adam (visited on 08/19/2024).

[2] "Vision: Processing Information," (), [Online]. Available: https://www.brainfacts.org/thinking-sensing-and-behaving/vision/2012/vision-processing-information (visited on 08/13/2024).

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[4] M. Rodriguez, “What Causes Red Eyes in Photos?” Barnet Dulaney Perkins Eye Center. (Feb. 15, 2018), [Online]. Available: https://www.goodeyes.com/eye-health/red-eyes-photos/ (visited on 08/13/2024).

[5] *History of the camera*, in *Wikipedia*, May 29, 2024. [Online]. Available: https://en.wikipedia.org/w/index.php?title=History_of_the_camera&oldid=1226239339 (visited on 08/19/2024).

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